

HOJAE LEE

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EXECUTIVE SUMMARY

Applied AI Scientist and Researcher bridging a rigorous mathematical foundation (B.S. in Mathematics, Korea University) with advanced clinical machine learning and multimodal data analysis. Published 21 peer-reviewed papers (SCI/SCIE) in top-tier medical journals within two years. Expertise spans building automated deep learning pipelines for medical imaging (Quantitative Coronary Angiography), engineering multi-agent LLM systems with Human-in-the-Loop validation, and mitigating LLM hallucinations in scientific datasets alongside MIT and University of Toronto researchers.

RESEARCH & PROFESSIONAL EXPERIENCE

Asan Medical Center, Biomedical Engineering Research Center *Seoul, South Korea | Mar 2025 – Present*
Researcher

- Build and deploy quantitative coronary angiography (QCA) analysis pipelines with clinician QA interfaces, significantly improving measurement consistency and reviewer efficiency.
- Prototype 3D vascular geometry reconstruction models derived from routine 2D angiographic sequences for advanced interventional planning contexts (TAVI).
- Develop semi-automated instrument (catheter/wire) tracking systems in angiographic videos utilizing foundation-model-assisted segmentation (SAM 2).
- Integrate structured clinical variables with imaging model outputs to generate clinician-facing explanations and summaries for robust decision support.

Independent Research Team (Private Academy) *Seoul, South Korea | Jan 2025 – Present*
ML Engineer

- Engineer a multi-agent LLM pipeline for curriculum-aligned mathematics problem generation, integrating Human-in-the-Loop (HIL) validation and Retrieval-Augmented Generation (RAG).
- Build an AI-assisted workflow for reading and grammar item generation and quality control, emphasizing choice-set consistency, distractor validity, and automated rationale generation.

University of Toronto & MIT *Toronto, Canada & Cambridge, MA | Dec 2023 – Jun 2024*
Joint AI Researcher (Fully Funded Scholar by MSIT)

- Collaborated with cross-functional research teams to mitigate data hallucination in Large Language Models (LLMs) processing complex scientific Q&A datasets.
- Implemented Parameter-Efficient Fine-Tuning (PEFT) and Soft Prompting techniques to optimize LLM performance while drastically reducing computational overhead (achieved compliant performance using <1% of fine-tuning parameters).
- Applied Knowledge Graph-based Retrieval-Augmented Generation (KG-RAG) to embed domain-specific physical science constraints into model outputs, significantly boosting factual accuracy.

M.S. Researcher

- Developed and externally validated ML risk prediction models for neurodegenerative, cardiovascular, and diabetic retinopathy outcomes in Type 2 Diabetes across two independent Korean cohorts.
- Standardized end-to-end clinical ML pipelines (data curation → feature engineering → model benchmarking → calibration/external validation) with clinician-interpretable XAI outputs (SHAP).
- Led nationwide time-series analyses of adolescent health behaviors (2005–2022), quantifying epidemiological trends and policy impacts (e.g., tax hikes, COVID-19).
- Authored/co-authored 21 peer-reviewed papers; built reproducible code-first reporting environments using Python/R.

MEDIAZEN

Seoul, South Korea | Jun 2021 – Jan 2022

NLP Researcher Intern

- Engineered text dataset preprocessing pipelines and generated structured input data using Python to train and deploy a commercial Korean chatbot system.

EDUCATION

University of Toronto

Toronto, Canada | Jan 2024 – Jul 2024

Visiting Student, Mechanical & Industrial Engineering

- Awarded full national fellowship by the Ministry of Science and ICT (IITP), South Korea.

Kyung Hee University

Seoul, South Korea | Mar 2023 – Feb 2025

M.S. in Pharmacy (Major: Pharmaceutical Big Data Analysis)

- Thesis: "Development of A Machine Learning-Based Prediction Algorithm for Suicidal Ideation in Adolescents and Independent Validation Across Three Worldwide Cohorts"

Korea University

Seoul, South Korea | Mar 2016 – Feb 2023

B.S. in Mathematics

TECHNICAL SKILLS

- **Machine Learning & AI:** PyTorch, TensorFlow, scikit-learn, LLM Fine-tuning (PEFT, sLLM pipelines), RAG, Prompt Engineering, Time-series modeling (LSTM, GRU), Survival/Mixed models, Interpretability (SHAP)
- **Medical Imaging & Vision:** QCA (segmentation, centerline, bifurcation extraction), 2D→3D angiography (TAVI), SAM 2, Catheter tracking
- **Programming Languages:** Python, R, MATLAB, C, JavaScript, SQL
- **Tools & Infrastructure:** Git, Docker, Linux, Jupyter, SAS, Human-in-the-Loop (HIL) systems
- **Languages:** Korean (Native), English (Professional Working Proficiency; TOEFL iBT 97)

SELECTED PUBLICATIONS

6 Co-first-author / 1 Corresponding-author among 21 peer-reviewed papers. Full list on [Google Scholar](#)

1. Yim Y, Lee JE, Son Y, Kim S, **Lee H**, et al. "Long-term trends in the prevalence of cardiovascular-kidney-metabolic syndrome in South Korea, 2011-2021." *Lancet Reg Health West Pac*. 2025 Jan.
2. Kim S, Kim H, Kim S, **Lee H**, et al. "Machine Learning-Based Prediction of Substance Use in Adolescents in Three Independent Worldwide Cohorts." *J Med Internet Res*. 2025 Feb.
3. Sang H*, **Lee H***, Park J, et al. "Machine Learning-Based Prediction of Neurodegenerative Disease in Patients With Type 2 Diabetes." *J Med Internet Res*. 2024 Oct. (*Co-first author)
4. Kattih M*, **Lee H***, Jo H*, et al. "National prevalence of atopic dermatitis in Korean adolescents from 2009 to 2022." *Sci Rep*. 2024 May. (*Co-first author)

5. Hong S*, Woo S*, ... **Lee H¶**, Yon DK¶. "National prevalence of smoking among adolescents at tobacco tax increase and COVID-19 pandemic in South Korea, 2005-2022." *Sci Rep.* 2024 Apr. (¶Corresponding author)

HONORS & AWARDS

- **Fellowship & Scholarship:** Ministry of Science and ICT (IITP) Fellowship (Dec 2023 – Jun 2024); Kyung Hee University Academic Scholarships (Spring 2023, Fall 2023, Spring 2024).
- **Research Grant:** AI Convergence Innovation Project Competition, Kyung Hee University (May 2023 – Dec 2023) — Early Prediction & Personalized Health Management Platform for Chronic Diseases.
- **Best Paper Award:** The 5th Engineering in Biomedical and Rehabilitation & Home Conference, Japan (Jun 2023).